# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $3\frac{1}{6}$ | Convert to improper fractions or borrow. |
| 2 | **(c)** 1 | Use the LCM of all denominators. |
| 3 | **(c)** 60 | Divide by 3, multiply by 5. |
| 4 | **(c)** $\frac{5}{8}$ | Fraction left $= 1 -$ fraction spent. |
| 5 | **(d)** $\frac{3}{8}$ | Take the average of the two fractions. |
| 6 | **(d)** 17 | Whole times denominator plus numerator. |
| 7 | **(b)** $\frac{3}{7}$ | Divide numerator and denominator by their HCF. |
| 8 | **(a)** 21 | Divide, then multiply. |
| 9 | **(d)** $2\frac{11}{12}$ | Convert to improper fractions or borrow. |
| 10 | **(b)** $\frac{3}{4}$ | Use the LCM of all denominators. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Convert to improper fractions (1); $3\frac{1}{4}$ (1). | 1+1 |
| 12 | Number $+$\frac{1}{2} of number $=$\frac{3}{2} of number (1); Number $= 18 \div$\frac{3}{2}$= 12$ (1). | 1+1 |
| 13 | Common denominator 24 (1); $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{8}$ (1). | 1+1 |
| 14 | Convert to improper fractions (1); $2\frac{1}{2}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ploughed $=$\frac{1}{4}$+$\frac{1}{3}$= \frac{7}{12}$ (1); Remaining $= 1 - \frac{7}{12}$ (1); $= \frac{5}{12}$ (1). | 1+1+1 |
| 16 | Mixed to improper: $\frac{5}{2}$ (1); Add $\frac{1}{3}$ (1); Subtract $\frac{1}{4}$: result $2\frac{7}{12}$ (1). | 1+1+1 |
| 17 | Filled $=$\frac{2}{3}$\times 6 = 4$ L (1); More needed $= 2$ L (1); Empty fraction $\frac{1}{3}$ (1). | 1+1+1 |
| | **OR:** Ploughed $=$\frac{2}{5}$+$\frac{1}{4}$= \frac{13}{20}$ (1); Remaining $= 1 - \frac{13}{20}$ (1); $= \frac{7}{20}$ (1). | 1+1+1 |
| 18 | Mixed to improper: $\frac{11}{3}$ (1); Add $\frac{1}{2}$ (1); Subtract $\frac{1}{6}$: result 4 (1). | 1+1+1 |
| | **OR:** Filled $=$\frac{3}{4}$\times 8 = 6$ L (1); More needed $= 2$ L (1); Empty fraction $\frac{1}{4}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Books: \frac{1}{4}$\times 120 = 30$ (1); Rest $= 90$ (1); Bag: 30 (1); Left $= 60$ (2). | 1+1+1+2 |
| | **OR:** Fraction added $=$\frac{3}{4}$-$\frac{1}{4}$= \frac{1}{2}$ (1); This part is 20 litres (2); Capacity $= 20 \div \frac{1}{2} = 40$ litres (2). | 1+2+2 |
| 20 | Books: \frac{1}{6}$\times 240 = 40$ (1); Rest $= 200$ (1); Bag: 80 (1); Left $= 120$ (2). | 1+1+1+2 |
| | **OR:** Fraction added $=$\frac{5}{6}$-$\frac{1}{3}$= \frac{1}{2}$ (1); This part is 15 litres (2); Capacity $= 15 \div \frac{1}{2} = 30$ litres (2). | 1+2+2 |

**Mistakes to avoid:**

- Adding the denominators as well: $\frac{1}{5} + \frac{2}{5} = \frac{3}{7}$ is wrong; the answer is $\frac{3}{5}$.
- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2\times2+1}{2} = \frac{5}{2}$.
- Comparing fractions by denominators only: a bigger denominator with the same numerator makes the fraction *smaller*.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 63 | Divide by 2, multiply by 7. |
| 2 | **(b)** $\frac{3}{5}$ | Fraction left $= 1 -$ fraction spent. |
| 3 | **(d)** $\frac{1}{2}$ | Take the average of the two fractions. |
| 4 | **(c)** 35 | Whole times denominator plus numerator. |
| 5 | **(a)** $\frac{5}{8}$ | Divide numerator and denominator by their HCF. |
| 6 | **(c)** 40 | Divide, then multiply. |
| 7 | **(d)** $3\frac{7}{12}$ | Convert to improper fractions or borrow. |
| 8 | **(c)** 1 | Use the LCM of all denominators. |
| 9 | **(d)** 48 | Divide by 5, multiply by 8. |
| 10 | **(d)** $\frac{7}{12}$ | Fraction left $= 1 -$ fraction spent. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Number $+\frac{1}{3}$ of number $=\frac{4}{3}$ of number (1); Number $= 20 \div \frac{4}{3} = 15$ (1). | 1+1 |
| 12 | Common denominator 10 (1); $\frac{7}{10}, \frac{3}{5}, \frac{1}{2}$ (1). | 1+1 |
| 13 | Convert to improper fractions (1); $3\frac{1}{10}$ (1). | 1+1 |
| 14 | Number $+\frac{2}{5}$ of number $=\frac{7}{5}$ of number (1); Number $= 28 \div \frac{7}{5} = 20$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ploughed $=\frac{1}{3}+\frac{1}{5} = \frac{8}{15}$ (1); Remaining $= 1 - \frac{8}{15}$ (1); $= \frac{7}{15}$ (1). | 1+1+1 |
| 16 | Mixed to improper: $\frac{7}{4}$ (1); Add $\frac{2}{5}$ (1); Subtract $\frac{1}{2}$: result $1\frac{13}{20}$ (1). | 1+1+1 |
| 17 | Filled $=\frac{3}{5} \times 10 = 6$ L (1); More needed $= 4$ L (1); Empty fraction $\frac{2}{5}$ (1). | 1+1+1 |
|  | **OR:** Ploughed $=\frac{3}{8}+\frac{1}{6} = \frac{13}{24}$ (1); Remaining $= 1 - \frac{13}{24}$ (1); $= \frac{11}{24}$ (1). | 1+1+1 |
| 18 | Mixed to improper: $\frac{13}{3}$ (1); Add $\frac{3}{4}$ (1); Subtract $\frac{1}{6}$: result $4\frac{11}{12}$ (1). | 1+1+1 |
|  | **OR:** Filled $=\frac{5}{6} \times 9 = \frac{15}{2}$ L (1); More needed $= \frac{3}{2}$ L (1); Empty fraction $\frac{1}{6}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Books: $\frac{1}{3} \times 180 = 60$ (1); Rest $= 120$ (1); Bag: 30 (1); Left $= 90$ (2). | 1+1+1+2 |
|  | **OR:** Fraction added $=\frac{4}{5}-\frac{2}{5} = \frac{2}{5}$ (1); This part is 18 litres (2); Capacity $= 18 \div \frac{2}{5} = 45$ litres (2). | 1+2+2 |
| 20 | Books: $\frac{2}{5} \times 300 = 120$ (1); Rest $= 180$ (1); Bag: 30 (1); Left $= 150$ (2). | 1+1+1+2 |
|  | **OR:** Fraction added $=\frac{2}{3}-\frac{1}{6} = \frac{1}{2}$ (1); This part is 25 litres (2); Capacity $= 25 \div \frac{1}{2} = 50$ litres (2). | 1+2+2 |

**Mistakes to avoid:**

- Adding the denominators as well: $\frac{1}{5} + \frac{2}{5} = \frac{3}{7}$ is wrong; the answer is $\frac{3}{5}$.
- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2\times2+1}{2} = \frac{5}{2}$.
- Comparing fractions by denominators only: a bigger denominator with the same numerator makes the fraction *smaller*.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $\frac{1}{2}$ | Take the average of the two fractions. |
| 2 | **(d)** 17 | Whole times denominator plus numerator. |
| 3 | **(b)** $\frac{4}{9}$ | Divide numerator and denominator by their HCF. |
| 4 | **(c)** 35 | Divide, then multiply. |
| 5 | **(a)** $3\frac{9}{10}$ | Convert to improper fractions or borrow. |
| 6 | **(b)** $1\frac{1}{5}$ | Use the LCM of all denominators. |
| 7 | **(b)** 63 | Divide by 4, multiply by 9. |
| 8 | **(a)** $\frac{4}{7}$ | Fraction left $= 1 -$ fraction spent. |
| 9 | **(a)** $\frac{1}{4}$ | Take the average of the two fractions. |
| 10 | **(c)** 25 | Whole times denominator plus numerator. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Common denominator 315 (1); $\frac{3}{5}$, $\frac{4}{7}$, $\frac{5}{9}$ (1). | 1+1 |
| 12 | Convert to improper fractions (1); $4\frac{3}{4}$ (1). | 1+1 |
| 13 | Number +\frac{3}{4} of number =\frac{7}{4} of number (1); Number $= 49 \div$ \frac{7}{4}$= 28$ (1). | 1+1 |
| 14 | Common denominator 24 (1); $\frac{11}{12}$, $\frac{7}{8}$, $\frac{5}{6}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ploughed =\frac{2}{7}+\frac{1}{2}= $\frac{11}{14}$ (1); Remaining $= 1 - \frac{11}{14}$ (1); $= \frac{3}{14}$ (1). | 1+1+1 |
| 16 | Mixed to improper: $\frac{13}{5}$ (1); Add $\frac{1}{2}$ (1); Subtract $\frac{1}{10}$: result 3 (1). | 1+1+1 |
| 17 | Filled =\frac{7}{8}×12 $= \frac{21}{2}$ L (1); More needed $= \frac{3}{2}$ L (1); Empty fraction $\frac{1}{8}$ (1). | 1+1+1 |
| | **OR:** Ploughed =\frac{1}{6}+\frac{2}{9}= $\frac{7}{18}$ (1); Remaining $= 1 - \frac{7}{18}$ (1); $= \frac{11}{18}$ (1). | 1+1+1 |
| 18 | Mixed to improper: $\frac{31}{6}$ (1); Add $\frac{2}{3}$ (1); Subtract $\frac{3}{4}$: result $5\frac{1}{12}$ (1). | 1+1+1 |
| | **OR:** Filled =\frac{2}{5}×5 $= 2$ L (1); More needed $= 3$ L (1); Empty fraction $\frac{3}{5}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Books: \frac{1}{8}×360 $= 45$ (1); Rest $= 315$ (1); Bag: 135 (1); Left $= 180$ (2). | 1+1+1+2 |
| | **OR:** Fraction added =\frac{7}{8}−\frac{3}{8}= $\frac{1}{2}$ (1); This part is 30 litres (2); Capacity $= 30 \div \frac{1}{2} = 60$ litres (2). | 1+2+2 |
| 20 | Books: \frac{3}{8}×480 $= 180$ (1); Rest $= 300$ (1); Bag: 60 (1); Left $= 240$ (2). | 1+1+1+2 |
| | **OR:** Fraction added =\frac{7}{10}−\frac{1}{5}= $\frac{1}{2}$ (1); This part is 24 litres (2); Capacity $= 24 \div \frac{1}{2} = 48$ litres (2). | 1+2+2 |

**Mistakes to avoid:**

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- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2\times2+1}{2} = \frac{5}{2}$.
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